

## SVM — Support Vector Machine

- **SVM** is a **supervised machine learning algorithm**.
- Mainly used for **classification**, but also supports **regression (SVR)**.

### 1. Hyperplane

SVM finds the **best boundary (hyperplane)** that separates different classes.

Example:



The | represents the decision boundary.

### 2. Maximum Margin

SVM doesn't just find any boundary. It finds the boundary with the **maximum margin** — the largest possible distance from the closest points of both classes.

### 3. Support Vectors

**Support vectors = data points closest to the decision boundary.**

They are important because they determine the position of the boundary.

Easy memory:

**SVM → Find the best boundary → maximize margin → closest points are support vectors.**

### 4. Linear vs Non-linear

- **Linear SVM** → data can be separated using a straight line/plane.
- **Non-linear SVM** → uses the **kernel trick** to handle complex boundaries.

### 5. Advantages

- Works well with **high-dimensional data**
  - Effective when classes are clearly separated
  - Can handle **non-linear classification** using kernels.
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## Kernel Trick — Notes

- **Linear SVM:** Cannot properly separate **non-linearly separable data**.
- **Kernel Trick:** Maps data **implicitly into a higher-dimensional space** where it can become linearly separable.
- It avoids explicitly creating all the higher-dimensional features, reducing computational cost.
- **RBF Kernel:** A popular kernel for non-linear data; uses the **distance/similarity between points** to create a non-linear boundary.
- **Polynomial Kernel:** Creates non-linear boundaries based on the selected **polynomial degree**.
- Different kernel choices and parameters can produce different results.

